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ChemE for Sustainability

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Assessing the feasibility of Corn on the Polylactic Acid Market





Introduction

The U.S. Environmental Protection Agency (EPA) states that the U.S. produces more than **40 million tons of plastic** annually. Of these 40 million tons, **less than 9% was recycled**. Once we dispose of these plastic bottles, bags, and packages from our daily lives, they do not simply disappear. Rather, they accumulate in landfills at dangerous rates.

The EPA also shares that in 2023 alone, the US sent **27 million tons of plastic waste to landfills**. This alarming number calls for the need to properly address plastic waste. It demonstrates that recycling initiatives today are severely impractical and poorly enforced and overall, plastic usage itself should be reduced.

Through effective policy and government intervention, recycling may become more standardized. However, the dangers of existing and future plastics will still remain. These **plastics will not biodegrade**, rather they will take centuries to breakdown into microplastics that will **accumulate in our oceans and environment**.

Rather than waiting to reach a point of no turning back (though some may argue we have already reached it) we must consider alternatives to the practices and the plastics we use today. Is there a **greener, more sustainable option** to the main plastics in use today: Polyethylene Terephthalate (PET), High-Density Polyethylene (HDPE), Low-Density Polyethylene (LDPE), Polyvinyl Chloride (PVC), Polypropylene (PP), Polystyrene (PS)?



Figures on Plastic Production & Waste

1960-2018 Data on Plastics in MSW by Weight (in thousands of U.S. tons)

Management Pathway	1960	1970	1980	1990	2000	2005	2010	2015	2017	2018
Generation	390	2,900	6,830	17,130	25,550	29,380	31,400	34,480	35,410	35,680
Recycled	-	-	20	370	1,480	1,780	2,500	3,120	3,000	3,090
Composted	-	-	-	-	-	-	-	-	-	-
Combustion with Energy Recovery	-	-	140	2,980	4,120	4,330	4,530	5,330	5,590	5,620
Landfilled	390	2,900	6,670	13,780	19,950	23,270	24,370	26,030	26,820	26,970

Figure 1. 1960- 2018 Data on Plastics in MSW by Weight (in thousands on U.S. tons)

Plastics: Material-Specific Data. US EPA.

<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data>.

Plastics Waste Management: 1960-2018

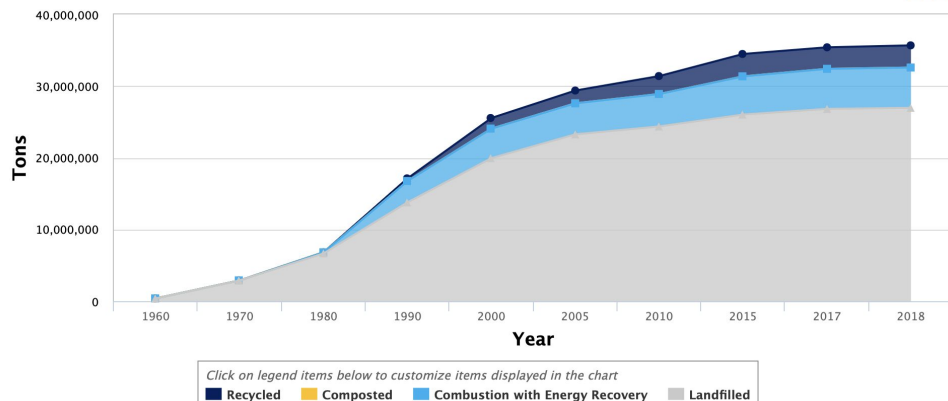


Figure 2. Plastic Waste Management 1960-2018

Plastics: Material-Specific Data. US EPA.

<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data>.

Which plastics could bio polyesters replace and how large are those markets?

Rigid packaging can include beverage bottles, thermoformed clamshells, and many food containers. The incumbent materials are PET and polystyrene, and some polypropylene in injection molded parts. Flexible packaging can include produce bags and liners. Polyhydroxyalkanoates can substitute for polyethylene or polypropylene when certified compostability is required. Fibers and textiles can include apparel and nonwovens: polyester dominates, while polylactic acid fibers are niche and growing in specific segments. Medical and specialty items can include resorbable implants, sutures, pins, and agricultural mulch films where biodegradation or resorption has actual value. Polyester fiber was about seventy one million metric tons in 2023 and about seventy eight million metric tons in 2024. PET bottles are about twenty six million metric tons. global plastics production was roughly four hundred to four hundred thirty six million metric tons in 2023. Bioplastics capacity in 2024 was about two point four seven million metric tons with actual production about one point four four million metric tons, which really shows meaningful headroom if the demand materializes.

Examples of items that can be made from bio polyesters:

Polylactic acid can be used for thermoformed food packages, cold cups and lids, coffee capsules, and 3D printing filament. It also appears in a lot of medical devices that benefit from resorption. Polyhydroxyalkanoates can be used for compostable straws and cutlery, compostable bags, and agricultural films. Some grades actually show faster biodegradation in certain test environments.

Market value snapshots for displaced polymers

Polyester fiber market value was on the order of one hundred to one hundred thirty billion dollars in 2024 (depending on the source and scope definition). PET bottles market value was about forty five to forty six billion dollars in 2024 with a volume near twenty six million metric tons. Polystyrene markets add several tens of billions of dollars globally.

PLA Production Through Corn:

The production of PLA from Corn involves 7 Crucial Steps and Reactions:

Corn $\xrightarrow{1}$ Starch $\xrightarrow{2}$ Glucose $\xrightarrow{3}$ Lactic Acid $\xrightarrow{4}$ Lactide $\xrightarrow{5}$ PLA

Namely, this Process Involves the Following Relevant Chemical Reactions and Conversions:

1. A ~72.3% loss from converting Corn to Corn Starch
2. $(C_6H_{10}O_5)_n + nH_2O \rightarrow n(C_6H_{12}O_6)$ - Starch to Glucose reaction
3. $C_6H_{12}O_6 \rightarrow 2CH_3CH(OH)COOH$ - Conversion of Glucose to Lactic Acid via Fermentation
4. 30% Lactic Acid to Lactide evaporation and reactivity loss
5. $\frac{72.065}{90.081} \approx 0.80 \rightarrow 80\%$ mass factor conversion to PLA - Lactide to PLA conversion

Investments and Demands to Satisfy PLA Creation:

Initial/Sample Case Analysis: 1,000kg Sample:

$$1,000\text{kg Dry Corn} \times 0.723 = 723\text{kg Starch}$$

$$\frac{\text{Starch Molar Mass}}{\text{Glucose Molar Mass}} = \frac{180.16}{162.1} \approx 1.11 \Rightarrow 723 \times 1.11 = 802.53\text{kg Glucose}$$

$$802.53 \times 1 = 802.53\text{kg Lactic Acid}$$

$$\frac{\text{Lactic Acid Molar Mass}}{\text{PLA (Repeating) Molar Mass}} = \frac{90.081}{72.065} \approx 0.80 \Rightarrow 802.53 \times 0.80 = \underline{642.024\text{kg PLA}}$$

Thus, a 1,000 kg sample of Corn will provide a theoretical yield of ~642 kg Polylactic Acid.

Accounting for a practical efficiency loss of 30% for unwanted byproducts, heat loss, and fermentation/purification un-optimizations results in a final yield % of **44.9%**

$$642.024 \times 0.7 = 449.4168\text{kg of PLA} \Longrightarrow \underline{\sim 44.94}$$



Investments and Demands to Satisfy PLA Creation

(2):

Global Demand Given Estimations:

First, let us note the Annual Polyester Production current/recent estimations≈ *71.1 million metric tons*

Using our estimated conversion percentage of 44.9%, we find the necessary quantity of corn to completely replace the PLA industry to be:

$$71.1 \times \frac{1}{0.449} = 158.352 \text{ million metric tons of corn}$$

With the global corn production projected to land at around 1.288 billion metric tons in the 2025-2026 year, this would leave the relative corn demand for the PLA industry at:

$$\frac{158.352}{1,288.58} = 0.1228887613. \text{ Or just over } 12\% \text{ of the global corn market.}$$

Or, in terms of expected land usage:

$$380,000,000 \times 0.1289 = 46,697,729 \text{ acres of land across the world}$$


Polyesters & Switching to Biopolyesters

Polyester is the third most widely used plastic, and it is **produced from crude oil** through the process of petrochemical refinement and polymerization. Its **tensile strength, durability**, and resistance to shrinking and stretching makes it an essential material across many industries. Additionally, polyesters such as PET are **economically favorable** as they are relatively inexpensive to manufacture. As there is existing infrastructure to recycle these plastics, polyesters such as PET have the potential to be recycled and reused instead of continuously produced.

The production of polyester requires non-renewable resources such as fossil fuels, which isn't feasible long term. Polyester is one of the more **carbon-intensive** materials in the textile industry as the production of one kilogram releases, on average, 14.2 kilograms of carbon dioxide. Though it has the potential to be recycled, polyester will not biodegrade on its own and requires extreme temperatures and pressures. Because industrial recycling is required and only 14% of the PET generated annually is recycled, overall polyester contributes to **significant environmental pollution and plastic waste**.

Looking at the alternative, biopolyesters are not produced from non-renewable resources such as fossil fuels. Rather they are **derived from sustainable resources** such as plants, algae, and microorganisms. This poses a significant advantage as it reduces the reliance on fossil fuels, thus **emitting less greenhouse gasses** in the production process. Additionally, as biopolyesters tend to be less durable, they can be used for single-use items meant to have a shorter lifespan.

While favorable in these aspects, biopolyesters **requires lots of land usage, irrigation systems and water, and fertilizer needs**. To displace polyesters, even marginally, these factors would cause **significant environmental strain**. Furthermore, the biopolyesters being less durable pose an issue for items that are multi-use and have longer lifespans. Looking at this from an economic standpoint, the more expensive and underperforming biopolyester is not enough to entice manufactures to make this switch from polyester. Investors would have to expect lots of **risk and price volatility**. Lastly, as with polyester, biopolyester is **not entirely biodegradable** in all environments, such as landfills. Therefore, it still requires industrial recycling infrastructure for complete and effective biodegradation, with a **heavy reliance on people choosing to properly recycle**.

Potential Next Steps and Areas of Improvement

A major hindrance in switching from polyesters to biopolyesters is the **poor durability and brittleness of the biopolyesters**. To increase the strength and impact resistance, methods such as nanoparticle and plasticizer incorporation can be used.

Nanoparticle incorporation enables **toughening and compatibilization of PLA** and other brittle biopolyesters through blending ductile polymers or elastomers like natural rubber. Through improving adhesion between phases, materials with desirable properties are created. The properly compatibilized PLA blends demonstrate improvements in strength, elongation, and impact resistance.

Plasticizers are typically used in materials like plastics or concrete to **increase durability and elasticity**. By weakening the bonds between polymer chains, the polymers can move more freely and create more flexible products. Plasticizers include naturally derived compounds like glycerol, epoxidized soybean oil, and castor oil.

Methods such as nanoparticle and plasticizer incorporation provide the **basis of improvement for the quality and desirability** of biopolyesters. Through this, its market can increase and slowly work to displace polyesters with a sustainable and greener alternative that preforms well.

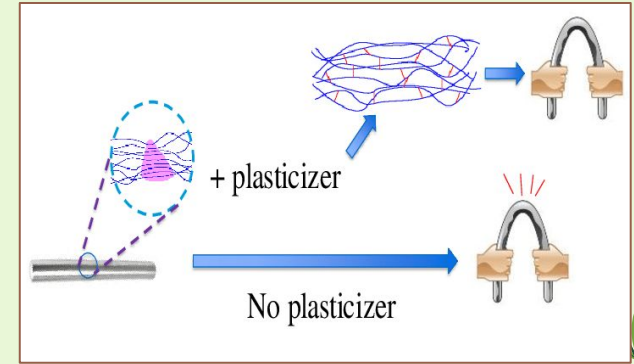


Figure 1. Illustration of the role of a plasticizer in a polymer matrix

Saabome, Samuel & Abdul-Malik, Salim Umar & Adu-Poku, Brolin & Yoon, Soongseok. (2020). Title: A Review on Plasticizers and Eco-Friendly Bioplasticizers: Biomass Sources and Market. International Journal of Engineering Research and. V9. 10.17577/IJERTV9IS050788.

Conclusion

The PLA market is one of the **fastest growing and promising fields** of plastic-manufacturing. With a 2030 projection to tople \$4 billion, PLA is expected to account for over **20% of the plastic market**. As such, developing a concrete and sustainable pipeline to produce PLA reliably is crucial in order to continue our global sustainability goals.

Biopolyesters and biodegradable polymers such as PLA offer a potential greener alternative to the current widely used plastics such as PET and PP. While promising, to displace the PET demand and replace with PLA with current methods, the amount of corn needed would be inefficient and **cause enormous environmental strain** due to the land, water, and fertilizer required.

Metric:	Direct Corn to PLA Efficiency	Total Corn to Replace Entire PLA Market Demand:	Total Land Mass Needed Given Modern Corn Growth Spacings
Units:	%	Million metric tons	Acres
Estimation:	44.9	158.352	46,697,729

Summary of Estimation:

While producing and marketing PLA at this scale may not be feasible today, **advancements in PLA yield** and bioplastics through nanoparticle and plasticizer incorporation offer **potential solutions**.

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